

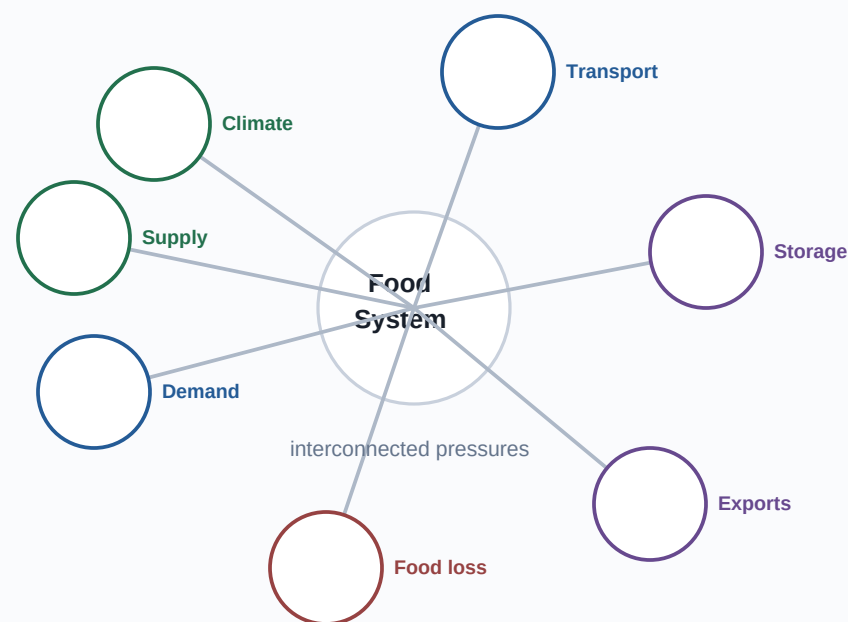
Systems Under Pressure

The Mathematics of Food Systems

"All models are wrong, but some are useful."

George Box

A standards-based assessment guide for reasoning, comparison, revision, and collaborative inquiry.



ASSESSMENT PHILOSOPHY

The rubric should teach what sophisticated mathematical thinking looks like.

Assessment prioritizes mathematical reasoning, model choice justification, comparison, critique, revision, graph-algebra-context connections, and visible collaborative thinking. It does not primarily reward aesthetics, editing, polish, or presentation smoothness.

Score 1-4 by standard.

Before grading, look for visible mathematical evidence.

The final format may vary. The mathematics must remain visible.

- Every graph has context, units, annotations, and realistic domain.
- Every model includes assumptions, limitations, and an explanation of why it fits.
- Every lens includes comparison across model types, versions, or assumptions.
- Every student contributes visible mathematical reasoning.
- r and R^2 are interpreted lightly as credibility and critique tools when relevant.
- Final communication centers reasoning, not production value.

NOT ASSESSED AS POLISH

Do not reward cinematic editing, visual effects, overproduction, or smooth performance as mathematical achievement. Clarity matters; performance polish is secondary.

STANDARD 1

Model Selection & Justification

Students explain why a model type fits the situation and what assumptions make it possible.

Look for discrete vs continuous arithmetic/geometric/exponential/polynomial choice variables, units, and assumptions why this model rather than another	1 Beginning Model choice is missing or mostly copied.	2 Developing Model choice is named but weakly justified.
	3 Proficient Model choice is justified with evidence and context.	4 Advanced Model choice is precise, comparative, and aware of tradeoffs.
STUDENT SELF-CHECK What evidence proves you are at the level you claim? Where is that evidence visible?	teacher notes / evidence codes _____ _____ _____	

STANDARD 2

Mathematical Reasoning

Students show accurate Algebra 2 thinking and explain the meaning of procedures, not only the answer.

Look for recursive and explicit formulas logarithmic threshold solving polynomial forms and operations correct units and interpretation	1 Beginning Reasoning is incomplete or conceptually confused.	2 Developing Some procedures are correct but explanations are thin.
STUDENT SELF-CHECK What evidence proves you are at the level you claim? Where is that evidence visible?	3 Proficient Reasoning is accurate and explained in context.	4 Advanced Reasoning is accurate, connected, and insightfully explained.
	teacher notes / evidence codes	

STANDARD 3

Comparison & Revision

Students compare related model versions and explain how changing assumptions changes predictions.

Look for multiple versions of models changed rates, dimensions, domains, or constraints recalculation after revision why the revision is stronger	1 Beginning Little evidence of comparison or revision.	2 Developing Comparison exists but is mostly surface level.
	3 Proficient Revision is explained with mathematical evidence.	4 Advanced Comparison and revision drive deeper understanding.
STUDENT SELF-CHECK What evidence proves you are at the level you claim? Where is that evidence visible?	teacher notes / evidence codes _____ _____ _____	

STANDARD 4

Graph-Algebra-Context Connections

Students translate among graphs, equations, tables, units, and real-world meaning.

Look for annotated graphs intercepts, extrema, multiplicity, end behavior realistic domains graph features interpreted physically	1 Beginning Representations are disconnected or unclear.	2 Developing Some connections are present but inconsistent.
STUDENT SELF-CHECK What evidence proves you are at the level you claim? Where is that evidence visible?	3 Proficient Graph, algebra, and context are clearly connected.	4 Advanced Connections are precise, layered, and used to critique
	teacher notes / evidence codes	

STANDARD 5

Interpretation, Limitations & r/R^2

Students interpret what the model reveals, what it misses, and how model fit affects credibility.

Look for assumptions and hidden variables correlation vs causation r and R^2 as lightweight fit evidence model breakdown and uncertainty	1 Beginning Limitations are missing or generic.	2 Developing Limitations are named but not well connected to evidence.
	3 Proficient Interpretation and limitations are clear and contextual.	4 Advanced Critique is nuanced, honest, and mathematically
STUDENT SELF-CHECK What evidence proves you are at the level you claim? Where is that evidence visible?	teacher notes / evidence codes	

STANDARD 6

Collaborative Mathematical Inquiry

The group investigates together while each student contributes visible mathematical thinking.

Look for shared reasoning across all lenses individual evidence snapshots authentic mathematical conversation every student speaks mathematics	1 Beginning Work appears divided or individual understanding is unclear.	2 Developing Collaboration is uneven or mechanically divided.
STUDENT SELF-CHECK What evidence proves you are at the level you claim? Where is that evidence visible?	3 Proficient Group inquiry is shared and individual evidence is visible.	4 Advanced Collaboration shows authentic critique, revision, and
	teacher notes / evidence codes	

Use evidence, not vibes.

For each standard, identify the artifact, page, graph, calculation, or

Model Selection

Level I claim: ____
Visible evidence:
Next revision:

Reasoning

Level I claim: ____
Visible evidence:
Next revision:

Comparison & Revision

Level I claim: ____
Visible evidence:
Next revision:

Graph-Algebra-Context

Level I claim: ____
Visible evidence:
Next revision:

Limitations & $r/R^\dagger$

Level I claim: ____
Visible evidence:
Next revision:

Collaborative Inquiry

Level I claim: ____
Visible evidence:
Next revision: